

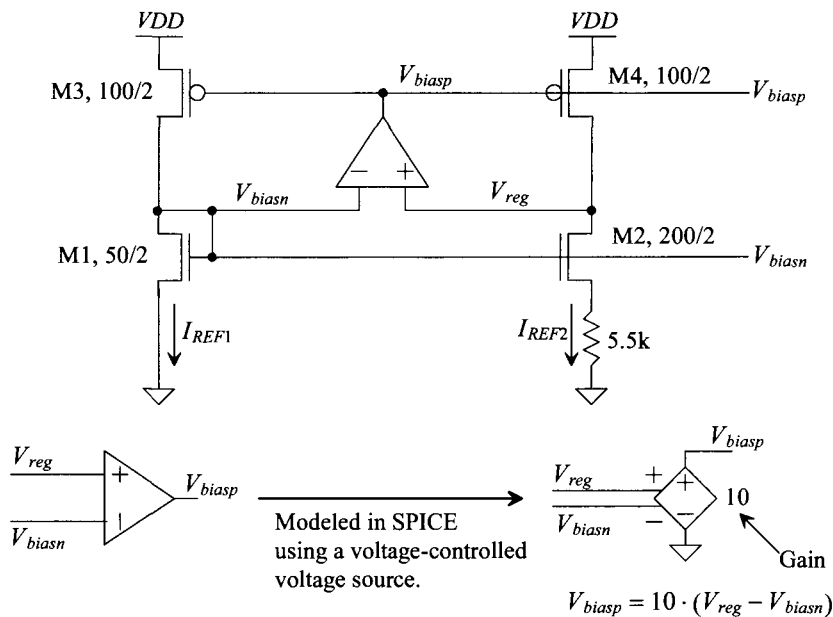


Figure 20.19 Increasing the output resistance of short-channel MOSFETs using feedback. The result, for the Beta-multiplier circuit, is better power supply sensitivity.

gate of M4 upwards, lowering the current it supplies and causing V_{reg} to drop back down. At the same time the gate of M3 is also increased, causing it to source less current. This causes a drop in V_{biasn} (the same as V_{reg} because of the symmetry as discussed in Fig. 20.1 or Ex. 20.3). Figure 20.20 shows how the reference current, in Fig. 20.19, changes with V_{DD} when the added amplifier has a gain of 10. We've lowered R to 5.5k to more accurately set the current and used, in the simulation, a voltage-controlled voltage source for the amplifier.

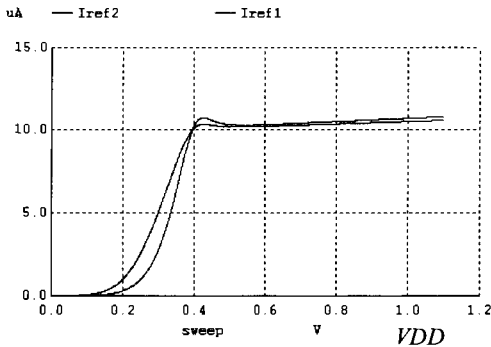


Figure 20.20 Improvement with the added amplifier.

Figure 20.21 shows a possible differential amplifier configuration. The PMOS devices form a current mirror while the NMOS devices will, when $V_{biasn} = V_{reg}$, simply mirror the current in M1. If $V_{biasn} \neq V_{reg}$ then the imbalance causes the amplifier output to swing up or down providing the desired action.

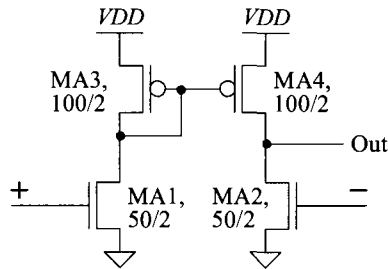
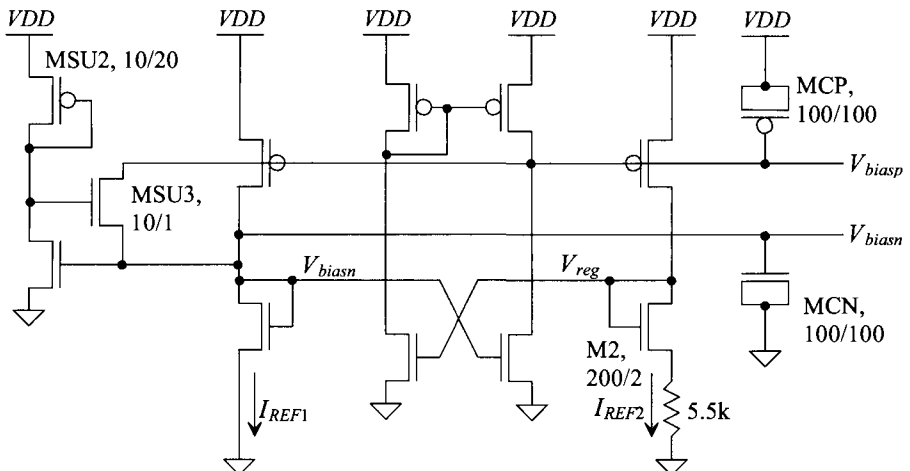


Figure 20.21 A possible implementation of the diff-amp in Fig. 20.19.

Figure 20.22 shows the improved current reference for use in a short-channel CMOS process (where devices have very low output resistances). The amplifier in Fig. 20.21 is placed in the reference, as indicated in Fig. 20.19. As with any feedback scheme, stability is of critical importance. As we'll see in the coming chapters, a feedback amplifier with only a single high-impedance node (meaning no diode-connected MOSFETs or MOSFET-source terminals are connected to the node) is straightforward to compensate (make stable). So that the reference has a single high-impedance node, that is V_{biasp} , we connected M2's gate to its drain (instead of to V_{biasn}). We get the same effect, as



All unlabeled NMOS are 50/2.
All unlabeled PMOS are 100/2.

Figure 20.22 Improved current reference for short-channel devices.
See also Sec. 23.1.3 for more information.

seen in Fig. 20.19 but the gain around the loop is reduced. To make the reference stable, we add capacitors (MCP and MCN) to the circuit. The high-impedance node is the critical point for adding the compensation capacitor (that is MCP is the critical capacitance). If the reference drives a significant number of MOSFETs (so these driven MOSFETs provide the load capacitance), then MCP and MCN may be excluded from the design. Figure 20.23 shows how the reference currents change with V_{DD} .

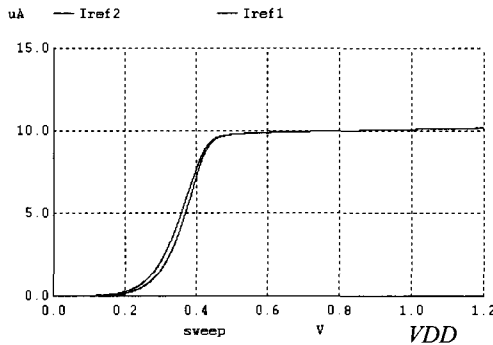
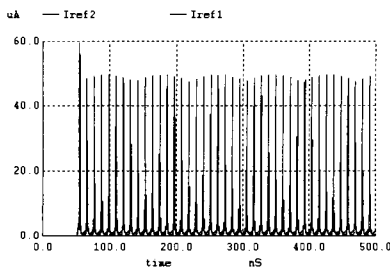


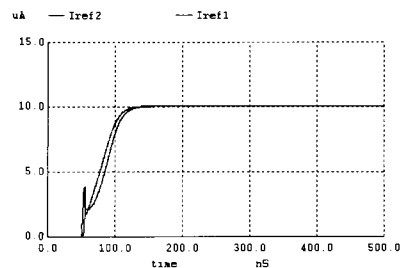
Figure 20.23 Variation of reference currents with V_{DD} for the circuit in Fig. 20.22.

An Important Note

It's extremely important to understand that together with the benefits of better power supply sensitivity are the undesired traits of feedback, that is, the potential for an unstable circuit. For example, Fig. 20.24a shows what happens to the reference currents if we remove MCP and MCN in Fig. 20.22 and then apply a step voltage to V_{DD} (that is, V_{DD} steps from 0 to 1 V at 50 ns in the simulation). Clearly, the reference is not stable and the currents oscillate. In (b) we do the same thing but with MCP and MCN present. The response shows first-order behavior, and oscillations are not present. We might think we are done with the reference and move on to the other analog circuit designs. However, further characterization of the current reference using simulations is warranted. For example, what happens if V_{DD} has a 50 mV squarewave signal coupled to it at 100 MHz? How does the reference behave with this signal (a squarewave V_{DD} that oscillates between 1 and 1.05 V at 100 MHz)? In general, no analog circuit has good power supply noise rejection at high frequencies. To remedy this, the power supply is decoupled (a large capacitor is placed from V_{DD} to ground to remove high-frequency noise).



(a) MCP and MCN not present



(b) MCP and MCN present

Figure 20.24 What happens when V_{DD} is pulsed from 0 to 1 at 50 ns.

20.1.5 Temperature Behavior

While we've been focusing on how the reference current changes with V_{DD} , it's also important to know how the current changes with temperature. Looking at the basic current mirror in Fig. 20.11a, note that if I_{REF} is a constant, independent of temperature, then the output current will also be independent of temperature. That's not to say that the MOSFET characteristics aren't changing because they are; they change at the same rate. Because they change together, the current mirror relationship is still valid and M2 mirrors the current in M1. Of course, if M1 is located at a different physical location than M2 and each is heated differently, a mismatch in the currents will result. *The point is that the temperature behavior of the reference current determines the temperature behavior of all mirrored currents.*

Using the short-channel CMOS parameters we know, from Ch. 9 (Fig. 9.30b), that the change in V_{THN} with temperature is $-0.6 \text{ mV}/^\circ\text{C}$ (for an NMOS device). For the PMOS device we can vary temperature and look at the change in V_{THP} . Using the mirror in Fig. 20.11 and looking at the V_{SG} at different temperatures results in the plots seen in Fig. 20.25. We see that it also varies at a rate of $-0.6 \text{ mV}/^\circ\text{C}$.

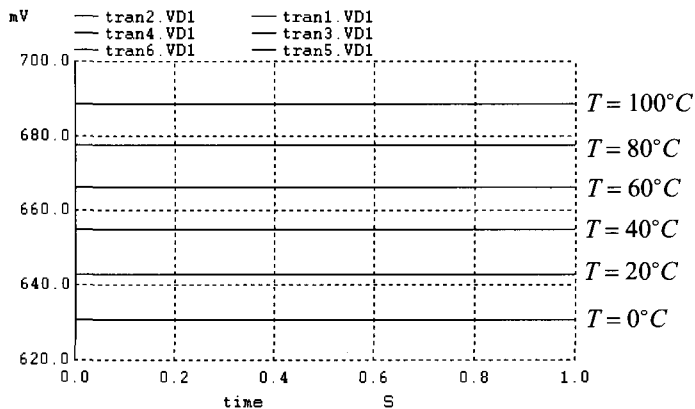


Figure 20.25 Variation in the reference gate voltage in the PMOS mirror seen in Fig. 20.11.

Resistor-MOSFET Reference Circuit

For a resistor-MOSFET reference circuit (Fig. 20.2), we can write (neglecting the MOSFET's finite output resistance)

$$I_{REF} = \frac{V_{DD} - V_{GS}}{R} \quad (20.26)$$

The change in I_{REF} with temperature is

$$\frac{\partial I_{REF}}{\partial T} = -\frac{V_{DD} - V_{GS}}{R^2} \cdot \frac{\partial R}{\partial T} - \frac{1}{R} \cdot \frac{\partial V_{GS}}{\partial T} \quad (20.27)$$

or

$$\frac{\partial I_{REF}}{\partial T} = -I_{REF} \cdot \frac{1}{R} \frac{\partial R}{\partial T} - \frac{1}{R} \cdot \frac{\partial V_{GS}}{\partial T} \quad (20.28)$$

We know that to write the reference current as a function of temperature we use

$$I_{REF}(T) = I_{REF}(T_0) \cdot (1 + TCI_{REF} \cdot (T - T_0)) \quad (20.29)$$

where the temperature coefficient of the reference current is

$$TCI_{REF} = \frac{1}{I_{REF}} \cdot \frac{\partial I_{REF}}{\partial T} \quad (20.30)$$

Rewriting Eq. (20.28) using Eq. (20.26) yields

$$TCI_{REF} = \frac{1}{I_{REF}} \cdot \frac{\partial I_{REF}}{\partial T} = -\frac{1}{R} \frac{\partial R}{\partial T} - \frac{1}{V_{DD} - V_{GS}} \frac{\partial V_{GS}}{\partial T} \quad (20.31)$$

Noting, again, that the temperature coefficient is not a constant but rather changes with temperature (even though this isn't indicated in Eq. (20.29)) just like a small-signal parameter, e.g., g_m , changes with the DC operating point.

Example 20.4

Determine the temperature behavior of the reference current in the mirror seen in Fig. 20.10. Verify the answer with SPICE.

If the temperature coefficient of the resistor, $TCR = \frac{1}{R} \frac{\partial R}{\partial T} = 2000 \text{ ppm}/^\circ\text{C}$ ($= 0.002$) and we assume after looking at Fig. 9.30 that $\frac{\partial V_{GS}}{\partial T} \approx \frac{\partial V_{THN}}{\partial T} \approx -0.6 \text{ mV}/^\circ\text{C}$ then, knowing the parameters used in Table 9.2 were the basis for the design of the current mirror in Fig. 20.10,

$$TCI_{REF} = -0.002/^\circ\text{C} - \frac{1}{1 - 0.35} \cdot (-0.6 \text{ mV}/^\circ\text{C}) \approx -1,000 \text{ ppm}/^\circ\text{C}$$

and so

$$I_{REF}(T) = 10 \cdot (1 - 0.001 \cdot (T - 27)) \mu\text{A}$$

At 100°C the reference current is $9.27 \mu\text{A}$ and at 0°C it's $10.27 \mu\text{A}$ (of course, the reference current is approximately $10 \mu\text{A}$ at 27°C). Figure 20.26 shows the SPICE simulation results. ■

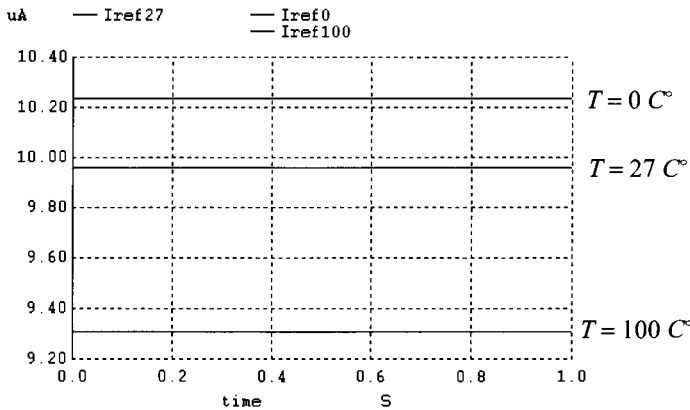


Figure 20.26 Example 20.4 showing the temperature behavior of the reference current in Fig. 20.10.

MOSFET-Only Reference Circuit

For the MOSFET-only bias circuit, Fig. 20.12, we can write the reference current as (see Eq. [20.16])

$$I_{REF} = \frac{VDD^2 - 2VDD \cdot (V_{THN} + V_{THP}) + (V_{THN} + V_{THP})^2}{\left(\sqrt{\frac{2L_3}{KP_p \cdot W_3}} + \sqrt{\frac{2L_1}{KP_n \cdot W_1}} \right)^2} \quad (20.32)$$

Knowing $KP_n = M \cdot KP_p$, where M is 3 for our long-channel process and 2 for our short-channel process (see Tables 9.1 and 9.2), we can write

$$I_{REF} = KP_n \cdot \frac{VDD^2 - 2VDD \cdot (V_{THN} + V_{THP}) + (V_{THN} + V_{THP})^2}{\left(\sqrt{\frac{M \cdot 2L_3}{W_3}} + \sqrt{\frac{2L_1}{W_1}} \right)^2} \quad (20.33)$$

or

$$\begin{aligned} \frac{\partial I_{REF}}{\partial T} = & \frac{\partial KP_n}{\partial T} \cdot \frac{VDD^2 - 2VDD \cdot (V_{THN} + V_{THP}) + (V_{THN} + V_{THP})^2}{\left(\sqrt{\frac{M \cdot 2L_3}{W_3}} + \sqrt{\frac{2L_1}{W_1}} \right)^2} + \\ & \frac{KP_n}{\left(\sqrt{\frac{M \cdot 2L_3}{W_3}} + \sqrt{\frac{2L_1}{W_1}} \right)^2} \cdot \left(-2VDD \cdot \left(\frac{\partial V_{THN}}{\partial T} + \frac{\partial V_{THP}}{\partial T} \right) + 2(V_{THN} + V_{THP}) \left(\frac{\partial V_{THN}}{\partial T} + \frac{\partial V_{THP}}{\partial T} \right) \right) \end{aligned} \quad (20.34)$$

Dividing both sides by Eq. (20.33) gives

$$\frac{1}{I_{REF}} \frac{\partial I_{REF}}{\partial T} = \frac{1}{KP_n} \frac{\partial KP_n}{\partial T} + \frac{2 \left(\frac{\partial V_{THN}}{\partial T} + \frac{\partial V_{THP}}{\partial T} \right) (V_{THN} + V_{THP} - VDD)}{VDD^2 - 2VDD \cdot (V_{THN} + V_{THP}) + (V_{THN} + V_{THP})^2} \quad (20.35)$$

noting, from Eq. (9.52), that the first term is simply $-1.5/T$.

Example 20.5

Determine the temperature behavior of the reference current in the MOSFET-only bias circuit seen in Fig. 20.13.

We note that both references in Fig. 20.13 have the same temperature behavior since Eq. (20.35) doesn't show a width or length dependence (there is a dependence in the reference current though). Figure 20.13 used the long-channel MOSFET parameters from Table 9.1 where

$$V_{THN} = 0.8, V_{THP} = 0.9, \frac{\partial V_{THN}}{\partial T} = -1 \text{ mV}/^\circ\text{C}, \frac{\partial V_{THP}}{\partial T} = -1.4 \text{ mV}/^\circ\text{C}$$

and $VDD = 5 \text{ V}$ with $I_{REF} = 13.5 \text{ }\mu\text{A}$ (measured in the simulation that generated Fig. 20.13) at $T = 300 \text{ K}$. Plugging the numbers into Eq. (20.35) gives

$$TCI_{REF} = \frac{1}{I_{REF}} \frac{\partial I_{REF}}{\partial T} = -\frac{1.5}{300} + \frac{2(-0.0024)(-3.3)}{25 - 17 + 2.89} = -0.005 + 0.00146 \approx -3,500 \text{ ppm}/^\circ\text{C}$$

and so $I_{REF}(T) = 13.5(1 - 0.0035(T - T_0)) \text{ }\mu\text{A}$. ■

Temperature Behavior of the Beta-Multiplier

To determine the temperature behavior of the Beta-multiplier, we take the derivative of Eq. (20.23) with respect to temperature

$$\frac{\partial I_{REF}}{\partial T} = \frac{-4}{R^3 K P_n \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \cdot \frac{\partial R}{\partial T} - \frac{2}{R^2 K P_n^2 \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \frac{\partial K P_n}{\partial T} \quad (20.36)$$

Dividing both sides by I_{REF} (Eq. [20.23]) gives

$$TCI_{REF} = \frac{1}{I_{REF}} \cdot \frac{\partial I_{REF}}{\partial T} = -2 \left(\frac{1}{R} \frac{\partial R}{\partial T} \right) - \frac{1}{K P_n} \frac{\partial K P_n}{\partial T} \quad (20.37)$$

Example 20.6

Determine the temperature behavior of the Beta-multiplier seen in Fig. 20.22.

If the temperature coefficient of the resistor, $TCR = \frac{1}{R} \frac{\partial R}{\partial T} = 2000 \text{ ppm}/^\circ\text{C}$ then

$$TCI_{REF} = -0.004 + \frac{1.5}{300} = 1000 \text{ ppm}/^\circ\text{C}$$

and so

$$I_{REF}(T) = 10 \cdot (1 + 0.001 \cdot (T - 27)) \mu\text{A}$$

The positive TC of the resistor subtracts from the negative TC of the mobility to stabilize the current reference. If the mobility change with temperature is complicated by velocity saturation effects, then simulations become invaluable to determine the actual temperature performance of the circuit. ■

Voltage Reference Using the Beta-Multiplier

Let's try designing the Beta-multiplier where the gate voltage of M1 (see Figs. 20.15 or 20.22) is a constant independent of temperature. In this type of design we aren't using this voltage directly to bias anything so we'll change the label from V_{biasn} to V_{REF} . Using Eqs. (20.22) and (20.23) gives

$$V_{REF} = V_{GS1} = \frac{2}{R \cdot K P_n \cdot \frac{W}{L}} \left(1 - \frac{1}{\sqrt{K}}\right) + V_{THN} \quad (20.38)$$

Taking the derivative with respect to temperature gives

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{THN}}{\partial T} - \frac{2}{R \cdot K P_n \cdot \frac{W}{L}} \left(1 - \frac{1}{\sqrt{K}}\right) \cdot \left(\frac{1}{R} \frac{\partial R}{\partial T} + \frac{1}{K P_n} \cdot \frac{\partial K P_n}{\partial T} \right) \quad (20.39)$$

If we want the change with temperature to be zero, $\partial V_{REF}/\partial T = 0$, then we can select the resistor based on

$$R = \frac{2}{\frac{\partial V_{THN}}{\partial T} \cdot K P_n \cdot \frac{W}{L}} \left(1 - \frac{1}{\sqrt{K}}\right) \cdot \left(\frac{1}{R} \frac{\partial R}{\partial T} + \frac{1}{K P_n} \cdot \frac{\partial K P_n}{\partial T} \right) \quad (20.40)$$

Using the parameters from Table 9.1 with $K = 4$ and a resistor temperature coefficient of 0.002 gives a resistor value of 5 k Ω . Figure 20.27 shows the SPICE simulation results. The variation is approximately 60 mV/100 $^\circ\text{C}$ or 600 $\mu\text{V}/^\circ\text{C}$.

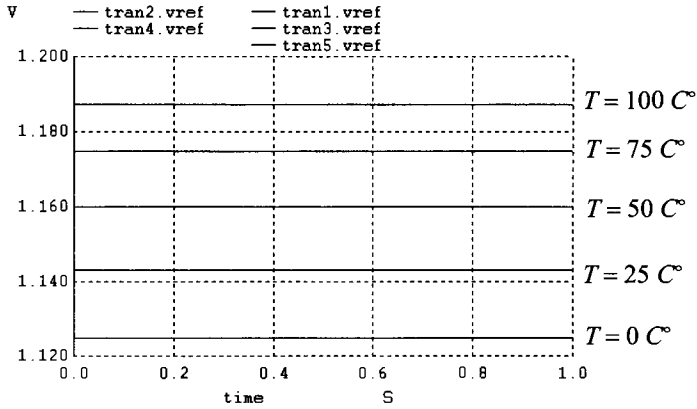


Figure 20.27 Temperature performance of a voltage reference using the Beta-multiplier.

20.1.6 Biasing in the Subthreshold Region

In some CMOS designs (for example, low-power designs) the MOSFETs may be operated in the weak or subthreshold regions. If a resistor is used to bias a current mirror, as seen in Fig. 20.10, its value can be very large. For example, if we use the short-channel process with $V_{DD} = 1$ V and a V_{SG} of approximately V_{THP} ($= 280$ mV) with an I_D of 10 nA, we need a resistor with a value of

$$I_{REF} = \frac{V_{DD} - V_{SG}}{R} \rightarrow R = 72 \text{ M}\Omega ! \quad (20.41)$$

Clearly, this isn't practical. To bias circuits for subthreshold operation, let's use the Beta-multiplier. From Eq. (9.17) we can write

$$V_{GS1} = nV_T \cdot \ln\left(\frac{I_{REF} \cdot L_1}{I_{D0} \cdot W_1}\right) + V_{THN} \quad (20.42)$$

and

$$V_{GS2} = nV_T \cdot \ln\left(\frac{I_{REF} \cdot L_1}{I_{D0} \cdot K \cdot W_1}\right) + V_{THN} \quad (20.43)$$

Knowing

$$I_{REF} = \frac{V_{GS1} - V_{GS2}}{R} \quad (20.44)$$

we get

$$I_{REF} = \frac{n \cdot V_T}{R} \cdot \ln K \quad (20.45)$$

or

$$R = \frac{n \cdot V_T}{I_{REF}} \cdot \ln K \quad (20.46)$$

Using the values above, that is, $K = 4$ and an I_{REF} of 10 nA, gives an R (assuming $n = 1$) of approximately 3.5 M Ω . Still large but much better than 72 M Ω .

20.2 Cascoding the Current Mirror

In this section we discuss increasing the output resistance of a current mirror so that it behaves more ideally. Figure 20.28 shows the problem with using a single MOSFET for a current source. Instead of being a constant, the output current, I_O , increases as the voltage across the current source, V_O , increases. We model this increase using the MOSFET's output resistance, r_o (see Table 9.2). If we can hold the drain-source voltage of the MOSFET constant, then the current doesn't vary. However, this requires a fixed V_O . To avoid this, we'll add circuitry in between the current-mirrored MOSFET and V_O .

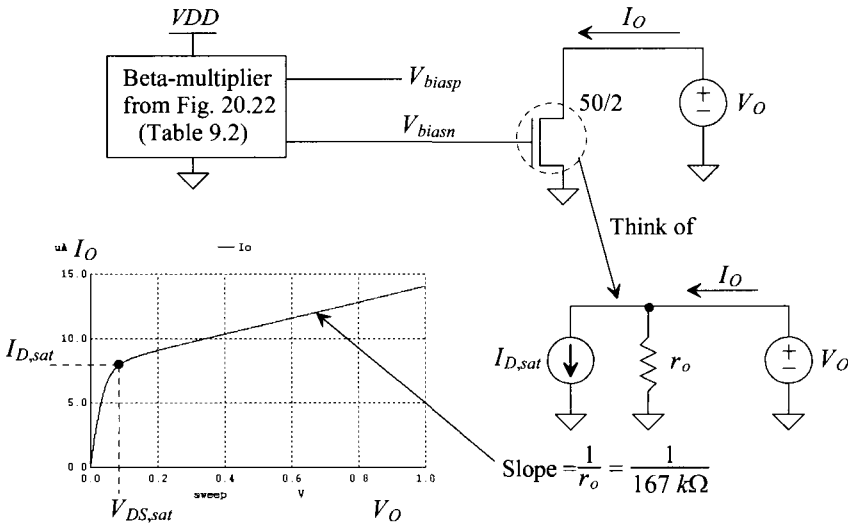


Figure 20.28 How the finite output resistance of the MOSFET affects the output current.

20.2.1 The Simple Cascode

The simple cascode current mirror made with M1–M4 is seen in Fig. 20.29. The name “cascode” is a vestige from the days of vacuum tubes when a common-cathode amplifier was cascaded (in series with) a common-grid amplifier. Drawing an analogy with MOSFETs, the cathode of a tube is equivalent to the source of a MOSFET. The grid of a tube corresponds to the gate of a MOSFET and the anode to the drain of the MOSFET.

It's important to realize that I_O is still determined by the gate-source voltages of M1 and M2. Changing the sizes of M3 and M4 simply changes the drain-source voltages of M1/M2, affecting the matching of their drain currents, as indicated by Eq. (20.10). Again, this (the gate-source voltages of M1/M2 determine the currents) is important to understand since, in the next several pages, we present different modifications to this basic cascode circuit. These modifications attempt to hold the drain source voltages of M1/M2 more constant to increase the current mirror's output resistance (make I_O change less with changes in V_O).

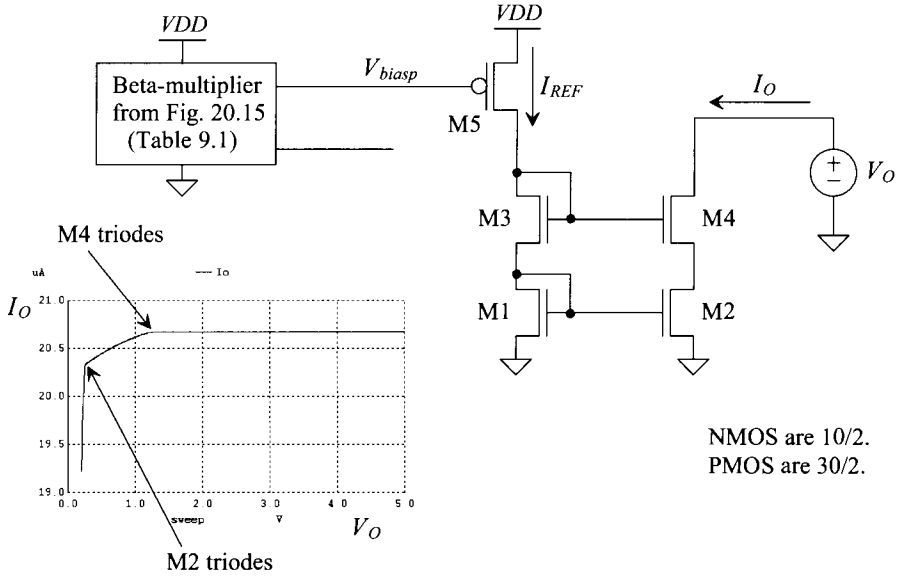


Figure 20.29 Biasing of the cascode current source and its operation.

DC Operation

The voltage on the gate of M4 in Fig. 20.29 is (remembering the actual values for these specific biasing conditions is given in Table 9.1)

$$2V_{GS} = 2(V_{DS,sat} + V_{THN}) \quad (20.47)$$

The voltage on the drain of M2, assuming that it is operating in the saturation region, is V_{GS} . The minimum voltage across the current source is then

$$V_{O,min} = V_{DS,sat4} + V_{GS} = 2V_{DS,sat} + V_{THN} \quad (20.48)$$

Plugging in the numbers from Table 9.1, we get $V_{O,min} = 1.3 \text{ V}$. Reviewing Fig. 20.29, we see that M4 starts to move into the triode region at this voltage.

To keep both M2 and M4 operating in the saturation region, we need only a $V_{DS,sat}$ across each one. In the next section we'll use this fact to design a *low-voltage cascode* current mirror.

Cascode Output Resistance

To estimate the output resistance of the cascode current mirror, let's apply a test voltage to the simplified cascode schematic seen in Fig. 20.30. We treat M1 and M3 as DC bias sources (AC grounds) and assume that the DC voltage on the drain of M4 is large enough to ensure that M2 and M4 are operating in the saturation. Further we draw the output resistances external to the devices. Note that M2's AC gate source voltage is zero and its drain voltage is $-v_{gs4}$. The resistance seen looking into the drain of M4 is

$$R_o = \frac{v_T}{i_T} \quad (20.49)$$

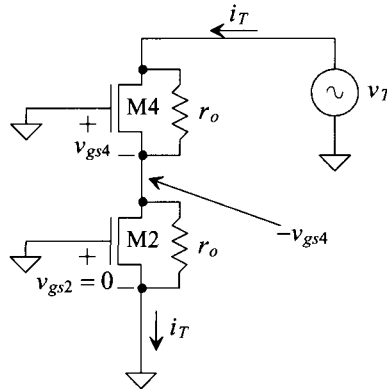


Figure 20.30 AC circuit used to determine the output resistance of a cascode current source.

Looking at Fig. 20.30, we can write

$$-v_{gs4} = i_T \cdot r_o \quad (20.50)$$

The current through M4 is then

$$i_T = g_m v_{gs4} + \frac{v_T - (-v_{gs4})}{r_o} \quad (20.51)$$

Substituting (20.50) into (20.51) gives

$$i_T = g_m (-i_T \cdot r_o) + \frac{v_T}{r_o} - i_T \quad (20.52)$$

Solving for the output resistance

$$R_o = (2 + g_m r_o) r_o \approx g_m r_o^2 \quad (20.53)$$

This result should be remembered because it will be referred to often. Using the numbers from Tables 9.1 and 9.2, we can tabulate the output resistance of cascode structures as seen in Table 20.1.

Table 20.1 Output resistances for cascode structures.

Cascode type	Long-channel process, Table 9.1, R_o	Short-channel process, Table 9.2, R_o
PMOS	2.4 G Ω	16.6 M Ω
NMOS	3.75 G Ω	4.2 M Ω

Notice the large difference (almost three orders of magnitude) between the output resistance of the short- and the long-channel devices. Again, this is the reason we need to use special circuit techniques when designing in short-channel processes.

Finally, note that the output resistance can be determined from a plot like the one seen in Fig. 20.29 (or Fig. 20.28) by plotting the reciprocal of the derivative of the output current.